Angles of similar triangles

1

a $x=28, y=89$ b $a=64, b=85$ c $p=71, q=38$ d $r=52, s=38$

2 127°

Sides of similar triangles

3

a $m=10, n=21$ b $g=30, f=5$ c $x=42, y=20$ d $a=9, b=66$

4

a NL b NM

Scale factor

5

a 3 b 3 c $\frac{3}{2}$ d $\frac{5}{4}$

6

a
i 2 ii $x=8$

b
i $\frac{8}{7}$ ii $a=32$

c
i 3 ii $x=12$

d
i 5 ii $y=5$

e
i $\frac{7}{2}$ ii $p=10$

f
i $\frac{8}{3}$ ii $q=56$

g
i $\frac{3}{2}$ ii $x=12$

h
i $\frac{2}{7}$ ii $m=4$

7

a True b True c False d True
e True

8 4

9 No solution provided.

10

a EF **b** $\angle EHG$

11

a 3 **b** $1:3$

12

a 0.5 **b** $2:1$

Applications

13 564 cm

14 23 m

15 $h = 15$

16 $h = 3.1$

17 $h = 1.4$

18 $h = 11.1$

19 $h = 4.8$

20 $d = 17.1$

21 $AB = 384 \text{ m}$

22 $w = 77$